

Article

Climate Policy Uncertainty and Renewable Energy Consumption: Multiple Insights from Digitization, Energy, and Innovation-Oriented Governance

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Abstract

This study applies the rolling window Granger causality method to explore the dynamic causal nexus between China's renewable energy consumption (REC) and climate policy uncertainty (CPU). From the perspectives of digitalization, energy development and technological innovation, empirical outcomes confirm that CPU exerts time-varying instead of static causal effects and steadily boosts REC expansion. There exists evident asymmetry in the reverse feedback of REC on CPU. Thriving renewable energy development reduces policy uncertainty by validating energy transition feasibility, while traditional energy interest-oriented policy revisions tend to heighten CPU. This study further verifies pronounced temporal heterogeneity in their causal interactions, providing fresh evidence for understanding the linkage between policy uncertainty and green growth. It finally proposes differentiated strategies involving short-term energy structure optimization and long-term climate institutional design to advance stable and resilient energy transition.

Keywords: climate policy risks; renewable energy; rolling window

1. Introduction

Fossil fuels continue to provide about 80% of the global primary energy consumption, despite the increasing utilization of renewable energy sources like wind and geothermal power [1,2]. Under the energy system dominated by fossil fuels, climate issues such as sea level rises, global warming, and ozone layer depletion are becoming increasingly serious [3,4]. These problems have significant negative impacts on energy supply, power generation efficiency, and the stability of energy infrastructure, thereby affecting energy production [5]. Moreover, climate change leads to declining agricultural productivity, long-term economic inequality and disruption of traditional energy supply chains [6–8]. Between 1998 and 2017, catastrophe connected to climate change cost the economy \$2.245 trillion [9]. Additionally, in the energy sector, it forces large Asian power companies to pay \$6.3 billion in material expenses each year [10]. To alleviate growing climate risks, many countries and organizations are paying more attention to formulate climate policies [11]. By 2030, the EU's European Green Deal aims to have 32% or more of the energy coming from renewable



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energy [12]. Australia published the Climate Change Act in 2023, and the government is funding the infrastructure for electric vehicles and the grid's integration of renewable energy [13]. Germany plans to generate 80% of the nation's electrical power from renewable energy by 2030 and attain carbon neutrality in its power supply system by 2035 [14]. The UK has launched a net zero strategy to generate all electricity from low-carbon energy by 2035 and to increase energy security by increasing domestic renewable energy production [15]. Almost 140 countries' climate policies have significantly contributed to making renewables cost-effective alongside conventional fossil-based energy facilities [16]. Renewable energy is expected to occupy about 20 percent of final energy consumption by 2030 [16]. However, the U.S. declared its intention to retreat from the Paris Agreement in 2017 but rejoined it in 2021, which undoubtedly increases global climate policy risks. Italy and many European countries are also beginning to consider adjusting their climate policies and reopening coal-fired power plants [17]. Argentina's withdrawal from the COP29 global climate talks has raised deep doubts about the direction of its climate policy. These events further increase the uncertainty expected in the future renewable energy market. Furthermore, digital development, technological innovation and energy transition interact closely in the current low-carbon context. Digital empowerment optimizes energy allocation efficiency, while continuous technological innovation accelerates the wide utilization of renewable energy [18]. The coordinated progress of digitalization and innovation can significantly alleviate policy uncertainty shocks, stabilize energy market expectations, and provide essential internal driving forces for renewable energy.

China is the global biggest developing state, which encounters daunting challenges in tackling energy transition and climate change. First, China has initially established a climate policy system. China has made a commitment to establishing a 1 + N climate policy structure, where "1" stands for the top-level design and guiding principle and the critical areas of the action plan are represented by "N" [19]. The government has promised to achieve carbon neutrality by 2060, which is reflected in national long-term growth plans [20]. Second, China's renewable energy industry develops rapidly. By the end of 2021, China had installed 330 GW (gigawatts) of wind power and 310 GW of solar power. The figures represent roughly 40% and 37% of the global totals for wind and solar power capacities, respectively [21]. By 2022, 2.7 trillion kilowatt-hours (kWh) of electricity has been generated using renewable energy, contributing to a 2.26 billion-ton reduction in carbon emissions [22]. The International Energy Agency (IEA) stated that China has great promise to improve renewable energy consumption to achieve environmental sustainability [23]. Third, China is actively pursuing energy transition. The preponderance of China's energy structure is currently coal-based, about 55.3% of energy consumption in 2023 [24]. China's installed renewable energy capacity grew by 302.2 GW in 2023, reaching a record high of 1450 GW, surpassing coal power generation capacity (1390 GW) [25]. Non-fossil energy is expected to increase from less than 20% in 2020 to more than 50% by 2050, with solar and wind each contributing 38% of electricity production [26]. The progress and growth of renewable energy is significant to China's power industry's large green transition [27].

This research study examines the dynamic time-varying causalities between climate policy uncertainty (CPU) and renewable energy consumption (REC) in China. In spite of an increasing number of studies on climate policy and energy transition, current research uses static models and overlooks the time-varying nature of policy impacts. Few investigations have formally tested the dynamic causal linkages between CPU and REC. This work uses a rolling window Granger causality technique to close this crucial gap, offering new and robust evidence on how climate policy risk dynamically shapes renewable energy consumption in China. The primary empirical findings are illustrated below. This study found that CPU promoted the growth of REC in the periods 2015:M09–2016:M03, 2017:M09–

2018:M03, and 2020:M01–2020:M04. This finding is in line with our expectation that increased climate policy risk will boost the application with renewable energy. Additionally, CPU adversely affected REC during the period 2014:M06–2015:M09, which was due to the technical constraints and specific climate policies of the time. Switching the causal direction of CPU and REC, the empirical findings indicate that REC had an adverse effect on CPU during the intervals 2009:M06–2009:M12 and 2017:M06–2021:M01, but this effect became positive in 2007:M03–2007:M09 and 2015:M01–2016:M03. Owing to advancements in renewable energy, the renewable energy industry is booming, while the conventional power industry is declining, which will bring about shifts in government policies. Overall, the causal association between CPU and REC is complex and unstable, and it only exists in a specific period.

Our research study conforms to the country's strategic orientation and provides a valuable contribution to the existing literature. First, the theoretical framework of this paper considers multiple pathways of REC effects on CPU, addressing the limitations of existing studies that usually consider the determinants of REC [28–30]. Our research work adds to the body of existing studies by employing rolling window-guided methods, which enable a detailed examination of time-varying causality, allowing us to identify such causality in distinct subperiods and thereby provide valuable insights into the structural changes within the CPU framework. Second, our findings can offer a fresh understanding of the connection between CPU and REC in China and advance the understanding of policy drivers among them. Compared with time-invariant assumption [31–33], this paper analyzes the development situation and specific policies in China in different periods and considers the impact on the process of renewable energy technological progress and policy adjustment. Third, by concentrating on understudied aspects of the influence of CPU on REC, this research study broadens the scope of current studies. This analysis takes a “consumption” approach, whereas many earlier studies have primarily examined how climate policy affects “investment” in renewable energy [34–36]. Actually, REC quantifies how much renewable energy supplements or replaces conventional fossil fuel energy in order to meet consumption trends and energy demands. Renewable energy investment concentrates on project development, cost-effectiveness analysis, and financial decisions.

The remainder of this study is structured as follows: Section 2 covers the relevant studies. Section 3 introduces the theoretical basis. Section 4 shows the data and method. Section 5 offers a detailed explanation of the empirical analysis. Section 6 presents discussion and conclusions.

2. Literature Review

2.1. China's Climate Policy and Renewable Energy Development

China's climate policy has a profound impact on development in national political, economic, social and other aspects [37]. Hurri [38] showed that China's climate policy is poised to exert a substantial influence on the global endeavors to mitigate climate change. Wu [19] found that China's climate governance is short of policy diversity, interest alignment and public participation, requiring changes such as enactment of national laws and optimization of policy mix. Huang and Zhu [39] argue that the climate policy of China has a huge influence on global carbon emission reduction, which is really encouraging for nations to cope with climate change. Chu et al. [40] argued that the climate policy stringency would encourage companies to make energy innovations, with the impact being more pronounced in areas with more regulations. Zhang et al. [41] noted that the change in climate policy increased urban climate risk in China and led to negative outcomes. Xiang and Gevelt [42] suggested that China's policymakers should proceed cautiously to ensure that their pursuit of global climate leadership does not lead to sacrifices in domestic

support for national policies. Zhang et al. [43] noted that the composite climate strategy will encourage GDP development and a clean change in the energy mix and could alleviate economic inequality to some degree.

Advancing renewable energy development is an inevitable option to achieve China's current climate policy goals and the energy transition. Ma and Wang [44] showed that energy policy always gives priority to "development" and emphasizes the importance of "new energy". Wang and Lei [22] found that renewable energy development exhibits a significant positive spatial correlation with carbon emission intensity across various provinces. Wang and Zhen [45] argued that China's photovoltaic development is fueled by the production sector and that wind energy production is propelled by a confluence of production and consumption dynamics. Yang et al. [46] showed that advancement in renewable energy can accelerate the energy transition, mediating technological improvements and green performance. Wei et al. [47] verified that stringent requirements for renewable portfolio standards are conducive to boosting the utilization efficiency of non-hydropower renewable energy installations across China. Teng et al. [21] pointed out that scaling up renewable energy output helps enhance provincial-level energy operational performance in China.

2.2. Climate Policy Risks and Renewable Energy Consumption

Current studies discuss the association from climate policy risks to renewable energy consumption; here we mainly describe four types of influence: positive, negative, heterogeneous and alleviating. First, the increase in CPU will boost REC. Shang et al. [48] found that CPU decreases the demand for traditional energy sources and therefore can promote renewable energy consumption by implementing identified climate policies. Liang et al. [49] showed that CPU has shown excellent capacity to estimate the variations in renewable energy. Crichton et al. [50] noted that REC grows by 2.64% to 4.43% per year after the implementation of climate policies and does not adversely affect economic growth. Tian et al. [51] explained that CPU plays a positive part in renewable energy progress and that a CPU drop has a greater impact than its increase. Guo et al. [52] showed that CPU has significantly impacted energy markets, which will catalyze the shift to renewable energy.

However, REC is able to alleviate CPU. Shang et al. [53] emphasized that the emission reduction effect of REC will help mitigate physical risks of climate change and reduce CPU. Zhang et al. [54] showed that REC can enhance the effectiveness of climate policy and that policymakers should develop relevant regulations to achieve REC goals. Dunbar and Treku [55] stressed that increasing REC is critical to meeting the global climate commitment and that investment in the energy transition should be increased. However, the shift towards renewable energy could potentially influence CPU. Aall et al. [56] argued that fast shifting between traditional energy and climate-related renewable energy systems could sharpen CPU. Kim and Park [57] found that climate disparity is made worse by the switch to renewable energy, which also has an impact on how well certain nations adapt to climate change.

In summary, this paper systematically sorts out extant studies concerning China's climate policy evolution and renewable energy advancement as well as the interactive relationship between climate policy uncertainty (CPU) and renewable energy consumption (REC). Through the collation of studies, several research gaps are identified. First, most existing studies predominantly focus on the one-way promoting effect of CPU on REC yet neglect the reverse feedback mechanism [48,51,56]. Specifically, the practical dilemmas and developmental constraints within the renewable energy sector may in turn interfere with climate policy design and trigger fluctuations in CPU, which has rarely been fully discussed. Second, relevant empirical analyses mostly adopt conventional static econometric tech-

niques, failing to capture the complex systemic association and dynamic evolutionary traits between the two variables [40,41]. In light of such research deficiencies, this study adopts the rolling window Granger causality method to identify their dynamic two-way causal nexus and distinct time-varying properties, which facilitates an in-depth understanding of how CPU drives the low-carbon energy transition. Furthermore, this research study constructs a targeted theoretical framework clarifying the intrinsic connection between China's CPU and REC, thereby providing fresh research perspectives and empirical supplements for subsequent studies on climate governance and green energy development.

3. Theoretical Basis

This paper builds a unified conceptual framework for explaining the bidirectional time-varying causal nexus between CPU and REC which integrates three mutually supportive and synergistic dimensions: digitization, energy system operation, and technological innovation. The three mechanisms do not function independently but jointly regulate the dynamic interactive process between CPU and REC, forming a complete and systematic theoretical logic. The detailed integrated mechanism analysis is presented as follows.

3.1. The Mechanism of Digitization

As the underlying technical support for energy transition and policy implementation, digitization can accurately capture climate policy dynamics and provide risk early warnings for market entities. This helps enterprises adjust their renewable energy investment strategies and reduce investment risks [58]. Meanwhile, digital technologies can optimize the production efficiency of renewable energy and solve its intermittency and volatility problems [59]. In addition, digital platforms can integrate market information, promote cross-regional allocation of renewable energy, and amplify the driving effect of CPU on REC [60]. In the process of REC feeding back to CPU, digitization can provide accurate consumption data. This gives policymakers a basis for making decisions and reduces the blindness of policy adjustments. Meanwhile, the digital-driven energy internet can attain the concerted development of renewable and conventional energy [61]. This eases the resistance of traditional energy interest groups and mitigates climate policy risks, laying a digital foundation for the stable operation of energy systems and innovative technology application.

3.2. The Mechanism of Energy

Supported by digital empowerment and technological innovation, the energy system serves as the core operational carrier for the interaction between CPU and REC. The increase in CPU strengthens the policy direction for the low-carbon energy transition, directly driving the growth of renewable energy consumption. Simultaneously, energy security concerns caused by policy risks will also motivate market entities to increase their renewable energy deployment and lessen reliance on traditional energy, further enhancing the positive driving effect of CPU on REC [62]. In terms of the feedback from REC to CPU, there is an asymmetric effect. Sustained growth of REC and prosperity of the renewable energy industry will verify the feasibility of the energy transition. This reduces the frequency of policy adjustments and mitigates climate policy risks. On the contrary, sluggish REC growth or obstacles to transition from traditional energy interests will increase policy uncertainty and exacerbate climate policy risks [63]. Notably, the realization of energy structure optimization and risk mitigation relies on digital resource allocation advantages and innovative technological breakthroughs, and in turn provides practical scenarios for digital upgrading and innovation iteration.

3.3. The Mechanism of Innovation

Driven by policy uncertainty and optimized energy system demands, technological innovation acts as the core driving force to strengthen the interactive linkage between CPU and REC and coordinate digital development and energy transition. Growing CPU will compel businesses and academic organizations to invest more in innovation in renewable energy [64]. The focus is on the R&D of core techniques like wind power, photovoltaic, and energy storage. Technological breakthroughs can raise the efficiency of renewable energy conversion, and model innovation can reduce investment and operation costs [65]. This makes renewable energy more competitive in the market and promotes the sustained growth of REC. In terms of REC feeding back to CPU, technological innovation can accelerate the speed and quality of the energy transition [66]. This enhances policymakers' confidence, reduces the frequency of policy adjustments, and lowers CPU. Additionally, innovation can promote the green transition of traditional energy and resolve conflicts of interest. This reduces the resistance to policy adjustments and weakens the negative feedback of REC on CPU. Complementary with digital technology and energy system optimization, innovation ultimately forms a synergistic system that jointly regulates the dynamic time-varying and asymmetric causal nexus between CPU and REC.

We suggest the following study hypotheses based on the theoretical examination of the three facets of innovation, energy, and digitalization mentioned above:

- H1.** CPU has significant Granger causal predictability for REC in China.
- H2.** REC also has significant causal predictive power for CPU.
- H3.** The dynamic link between CPU and REC has significant time-varying characteristics.

4. Data and Methodology

4.1. Data Source and Description

The sample duration selected in this study is from 2000:M1 to 2024:M8. This paper uses two variables for testing the two-way causal connection from CPU to REC. The first variable is CPU. It is measured by China's climate policy uncertainty index, which is a good explanation for climate policy volatility, and there have been a lot of studies on its application in energy economics [67]. The International Energy Transition Society's (IEFN) Energy Finance Committee website is the source of the CPU statistics. The second variable is REC. The monthly renewable energy generation, which encompasses solar, hydro, wind, and other energy sources, is utilized as the measurement variable [68,69]. A billion kilowatt-hours (kWh) is the unit of measurement. Because there is a shortage of full long-term monthly consumption-end statistical data, this study selects renewable energy generation as a credible proxy for REC, a common empirical treatment recognized in relevant studies [70]. Although generation data can well capture the overall variation characteristics of renewable energy utilization, this method still has inherent biases, including power transmission losses, wind/solar power curtailment and unmeasured intermediate energy circulation differences. Therefore, it is suggested that future research integrate more accurate, high-frequency actual consumption indicators to eliminate such estimation biases, verify the robustness of research results and extend related research insights. The National Bureau of Statistics of China is the source of the REC statistics. We apply a logarithmic transformation to REC, and then the differences between data can be reduced, making data analysis and modeling easier.

4.2. Bootstrap Full-Sample Causality Test

Stationarity is a prerequisite for Granger causality analysis. Existing simulations prove that RB approaches are more reliable than traditional asymptotic tests even without cointegration [71]. Shukur and Mantalos [72] also confirm that corrected LR tests perform better in small samples. We thus employ RB-based LR tests to explore the causality between China's CPU and REC, with the VAR(p) model specified as follows:

$$\begin{bmatrix} CPR_{1t} \\ REC_{2t} \end{bmatrix} = \begin{bmatrix} \varphi_{10} \\ \varphi_{20} \end{bmatrix} + \begin{bmatrix} \varphi_{11}(L)\varphi_{12}(L) \\ \varphi_{21}(L)\varphi_{22}(L) \end{bmatrix} \begin{bmatrix} CPR_{1t} \\ REC_{2t} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}, \quad t = 1, 2, \dots, T \quad (1)$$

$\varepsilon_t = (\varepsilon_{1t}, \varepsilon_{2t})'$ being 0 means an autonomous, random noise procedure with a non-singular covariance matrix Σ . In this analysis, we use the Schwarz Information Criterion (SIC) to ascertain the ideal lag order p . $\phi_{ijq}(L) = \sum_{k=1}^{p+1} \phi_{ij,k} L^k$, $i, j = 1, 2$, and the lag term L is specified to be $L^k x_t = x_{t-k}$.

By Equation (1), the assumption that China's CPU does not Granger-induce REC is verified by exerting the constraint $\varphi_{12,k} = 0$ for $k = 1, 2, \dots, p$. REC not Granger-causing Chinese CPU is examined under the constraint $\varphi_{21,k} = 0$ for $k = 1, 2, \dots, p$. When we reject the first hypothesis $\varphi_{12,k} = 0$ for $k = 1, 2, \dots, p$, a substantial causal connection exists from China's CPU to REC. This suggests that CPU can anticipate changes in REC. Likewise, when we reject the next hypothesis, $\varphi_{21,k} = 0$ for $k = 1, 2, \dots, p$, CPU is driven by changes in REC. Given that CPU affects REC, a strategic reduction in CPU by the government could help to control REC.

4.3. Stability Test of Parameters

Conventional full-sample Granger causality tests assume constant VAR model parameters over time, which may be invalid amid time-series structural changes. To address parameter instability, this study adopts Exponential-F (Exp-F), Supremum-F (Sup-F) and Mean-F tests to evaluate model stability with 15% trimming. For time-series observations $t = 1, 2, \dots, n$, parameters differ before and after the potential break point i . The F_i statistic is used to test the absence of structural breaks. Within the determined interval $[i_1, i_2]$, relevant calculations are presented as follows:

$$\text{Sup} - F = \sup_{i_1 \leq i \leq i_2} F_i \quad (2)$$

$$\text{Mean} - F = \frac{1}{i_2 - i_1 + 1} \sum_{i=i_1}^{i_2} F_i \quad (3)$$

$$\text{Exp} - F = \ln \left(\frac{1}{i_2 - i_1 + 1} \sum_{i=i_1}^{i_2} \exp \left(\frac{1}{2} F_i \right) \right) \quad (4)$$

4.4. Rolling Window Subsample Causality Test

This study utilizes the rolling window method to capture the time-varying nexus between CPU and REC, which conducts recursive estimation on fixed-length sliding subsamples across the whole period [73] and effectively captures dynamic variations and structural changes overlooked by full-sample estimation [74]; different from TVP-VAR models with strict parameter constraints and complicated operations, this method is more concise, transparent and free of rigid prior assumptions, thus being widely used in energy economics to track gradual dynamic trends [23], and it divides the full sample into multiple subsamples to capture time-varying causal nexus between variables.

Next, every subsample undergoes the RB-based modified-LR causality technique. Potential alterations in causal relationships from China's CPU to REC are identifiable intuitively through the computation of bootstrap p -values for measured LR statistics across rolling $T - l$ subsamples. T is the full sample, and l is rolling window size. The bootstrap estimation's mean was obtained from the formula $N_b^{-1} \sum_{k=1}^p \hat{\phi}_{21,k}^*$, with N_b signifying that the count of bootstrap iterations is used to determine the influence of China's CPU on REC. Likewise, the effect of REC on China's CPU was derived from the expression $N_b^{-1} \sum_{k=1}^p \hat{\phi}_{12,k}^*$. $\hat{\phi}_{21,k}^*$ and $\hat{\phi}_{12,k}^*$ represent bootstrap estimation produced by the systems in Equation (1).

This study sets the rolling window length at 24 months. The bootstrap method is further adopted to improve estimation accuracy and mitigate biases caused by limited window size. This window length is selected for three reasons: it matches the short-to-medium energy market cycle and policy adjustment cycle; it balances subsample quantity and estimation stability, avoiding inaccurate results from overly short windows and lagged structural change detection from excessively long windows; it is also a mainstream choice in existing energy and environmental economics time-varying research, facilitating consistent comparison with prior studies [75].

5. Results

5.1. Descriptive Statistic

Descriptive analyses are shown in Table 1. We can observe that the means of CPU and REC are 1.862 and 2.868. Additionally, the skewness of CPU is 0.298, a value greater than 0, suggesting that the data distribution exhibits right-skewedness in the tails. The skewness of REC is -0.200 , suggesting the distribution is skewed to the left in the tails. Every variable's kurtosis value is less than 3, suggesting that the data distribution is platykurtic, which is flatter than the normal distribution. Based on the results of the Jarque–Bera test, CPU and REC follow non-normal distributions.

Table 1. Summary statistics.

Variable	Mean	Median	Max	Min	SD	Skewness	Kurtosis	Jarque–Bera
CPU	1.862	1.838	4.121	0.291	0.805	0.298	2.321	10.075 ***
REC	2.868	2.887	3.488	2.079	0.362	-0.200	1.807	19.507 ***

Note: SD denotes standard deviation. *** denotes significance at the 1% level. The null hypothesis of the Jarque–Bera test indicates normal distribution.

Figures 1 and 2 illustrate fluctuations in CPU and REC over the sample time period. Figure 1 shows that the value of the CPU indicator has risen significantly since 2009. The failure of the Copenhagen climate talks in 2009 and the vacillation of the international community's attitude towards climate policy spread to China, leaving CPU relatively high. In March 2024, CPU even surpassed 4, as the COP28 agreed on an accelerated climate action package, prompting China to re-examine its adjusted National Decision Contributions (NDCs). Figure 2 shows that although REC fluctuates moderately up and down, it generally shows an upward trend. Since 2014, volatility has narrowed compared with the previous period. China implemented a dual energy consumption control policy to control overall energy consumption.

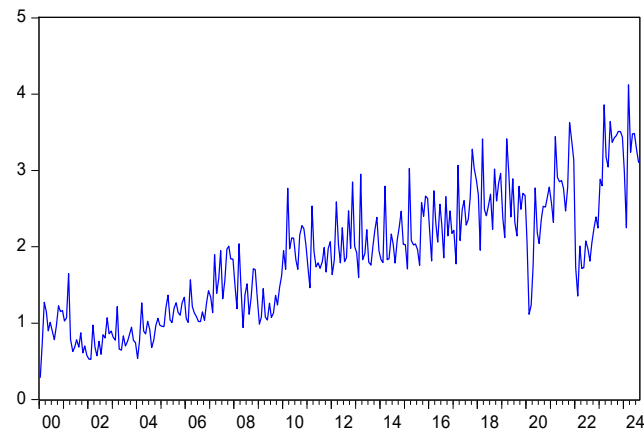


Figure 1. The trend of CPU. Note: The figure plots the monthly time series of the CPU index from 2000:M1 to 2024:M8. Higher values indicate greater uncertainty.

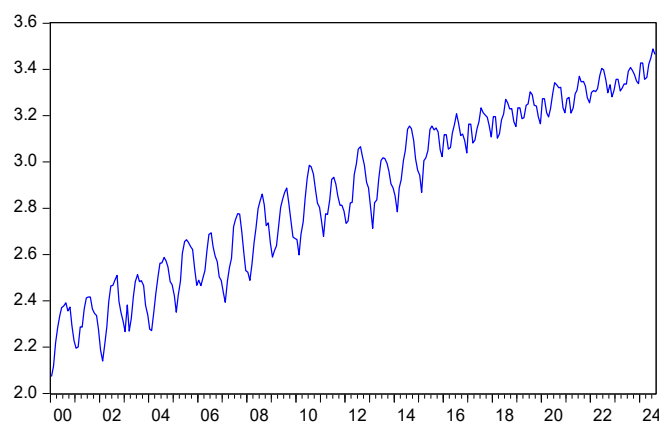


Figure 2. The trend of REC. Note: The figure plots the monthly time series of REC from 2000:M1 to 2024:M8. Higher values indicate greater consumption.

5.2. Full-Sample Granger Test

We first conducted a unit root test on the variables CPU and REC, and the ADF test showed that following first-order differencing, the variables are stable. Table 2 displays the unit root test's results, while Table 3 and Figure 3 provide the diagnostic test results of the VAR model. Therefore, we built a VAR model on this basis to estimate the full-sample causality using the two factors. The construction formula of the binary VAR model containing CPU and REC is shown in Equation (1). After verification, the determination of the optimal lag length for CPU and REC points to 4. The outcomes are displayed in Table 4. The null hypothesis, according to which CPU does not Granger-induce REC, is rejected. Similarly, the hypothesis whereby REC does not Granger-cause CPU is also rejected. The analysis reveals a Granger causal association between the variables in question. Hence, H1 and H2 can be validated. However, the full-sample test only reflects the average association and cannot capture the heterogeneous features across different time intervals.

Table 2. Unit root test.

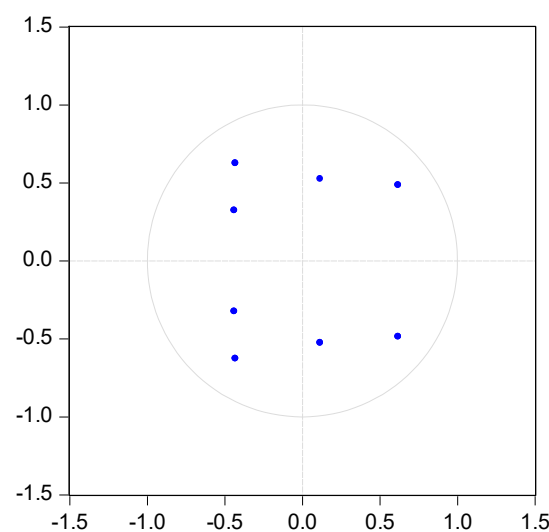
Variables	Augmented Dickey–Fuller (ADF)	
	Level	First Difference
CPU	−1.063	−5.535 ***
REC	−0.610	−7.936 ***

Note: *** denotes significance at the 1% level. The null hypothesis of the ADF test assumes the presence of a unit root.

Table 3. VAR diagnostics tests.

Test	Statistics	<i>p</i> -Value	5% Confidence Level	Explanation
Cointegration Test (Trace Statistic)	61.294	-	$r = 0$	Supports the use of VAR
Cointegration Test (Max-Eigenvalue)	33.643	-	$r = 0$	No cointegration
VAR Stability (Inverse Roots)	-	-	All roots < 1	VAR is stable
White Heteroskedasticity Test	190.571	0.198	Failed to reject H0	No heteroskedasticity
LM Autocorrelation Test (Lag 1)	19.239	0.316	Failed to reject H0	No serial correlation in residuals
LM Autocorrelation Test (Lag 2)	17.394	0.249	Failed to reject H0	No serial correlation in residuals
LM Autocorrelation Test (Lag 3)	16.847	0.271	Failed to reject H0	No serial correlation in residuals
LM Autocorrelation Test (Lag 4)	13.252	0.142	Failed to reject H0	No serial correlation in residuals

Note: Unit root tests indicate that both variables become stationary at the first difference; hence this study constructs a first-differenced VAR model. Given the absence of cointegration relationships, the VAR specification is preferred over the VECM. All inverse characteristic roots lie within the unit circle, verifying the stability of the established VAR system (see Figure 2). Moreover, LM autocorrelation tests fail to reject the null hypothesis, confirming that the model residuals are free from serial correlation and that the model setting is reasonable.

**Figure 3.** Inverse roots of the characteristic polynomial. Note: All inverse characteristic roots fall inside the unit circle, which verifies the stability of the constructed VAR model.**Table 4.** Full-sample Granger causality tests.

Tests	H ₀ : CPU Does Not Granger-Cause REC		H ₀ : REC Does Not Granger-Cause CPU	
	Statistics	<i>p</i> -Value	Statistics	<i>p</i> -Value
Bootstrap	13.446 ***	0.004	18.010 ***	0.002
LR test				

Note: *** indicates significance at the 1% level. When we reject the null hypothesis, Granger causality between variables exists.

Furthermore, to strengthen robustness, two control variables are incorporated in this study: China's economic policy uncertainty index (EPU) and the global Brent crude oil price (OP). EPU can isolate general policy shocks from climate policy-specific risks, and OP reflects energy market fluctuations [76,77]. All control variables adopt monthly frequency, consistent with the core variables' time dimension. After incorporating the above control variables and conducting robustness tests, Table 5 shows that the dynamic correlation between CPU and REC remains in line with the original results. This confirms that this paper's main conclusion is robust.

Table 5. Addition of control variables.

Model	Control Variables	CPU→REC	REC→CPU
(1)	EPU	15.5356 ***	21.5245 ***
(2)	EPU, OP	16.6035 ***	23.0361 ***

Note: *** denotes significance at the 1% level. “→” denotes the direction of causality.

Even though Table 4 clearly indicates causalities between the variables, the full-sample method assumes constant parameters over time. In reality, external shocks can affect variable interactions, causing structural changes and deviations in their causal relationships [78]. Compared with the conventional full-sample model, the rolling window method can effectively identify changes in the direction and strength of causality across different subperiods, revealing time-varying characteristics that cannot be captured by the full-sample test [79]. The findings suggest that the promoting effect of CPU on REC is significantly stronger in some periods but weaker in others and that the feedback effect of REC on CPU also exhibits distinct, phased differences. This indicates that using only the static full-sample test may obscure the true dynamic interaction between the variables, while the rolling window method offers a more thorough grasp of the complex causal mechanism between CPU and REC.

5.3. Parameter Stability Test

There is a default assumption in a large number of studies that there is no structural shift in the time sequence, and the causality during the sample period is directional [80]. However, when considering structural shifts, the VAR model’s parameters calculated using the whole sample CPU and REC will change over time. Hence, the premise of a single causal link and constant parameters in the full-sample causality test is obviously undependable. To tackle this problem, this article conducts parameter stability tests to ascertain if there are structural transitions. We assess the time stability of the parameters in the VAR model by using the Sup-F, Mean-F, and Exp-F tests. Table 6 lists the results. The initial line presents the Sup-F test outcomes, indicating a singular abrupt shift in China’s CPU, REC, and VAR system with a 1% confidence threshold. The outcomes of Mean-F and Exp-F are displayed separately on the two lines below; they indicate that the VAR equation constructed by CPU and REC might evolve incrementally over time. In summary, these results provide strong evidence to demonstrate the short-term instability of VAR model parameters using full-sample estimation. We can conclude from the stability test results that those indications of structural transformations make the model established with full-sample data unstable. Therefore, taking structural modifications into account, the rolling window test is necessary to re-examine the connection from CPU to REC.

Table 6. Parameter stability tests.

	CPU Equation		REC Equation		VAR System	
	Statistics	<i>p</i> -Value	Statistics	<i>p</i> -Value	Statistics	<i>p</i> -Value
Sup-F	32.287 ***	0.007	39.607 ***	0.002	52.413 ***	0.001
Mean-F	14.369 *	0.052	15.737 **	0.016	23.956 ***	0.008
Exp-F	12.510 **	0.044	15.934 **	0.021	22.337 ***	0.001

Note: We use 1000 bootstrap iterations to get *p*-values. *, ** and *** denote significance at 10%, 5% and 1%, respectively. Sup-F, Mean-F and Exp-F hold the assumption that the parameter has one-time sharp breaks and follows random walk, respectively.

5.4. Rolling Window Analysis

Figure 4 presents the rolling bootstrap *p*-values corresponding to the null hypothesis that climate policy uncertainty (CPU) has no Granger causal influence on renewable energy

consumption (REC), and the blue line in Figure 5 depicts the estimated values of bootstrap-based LR statistics. In this empirical analysis, we set the critical significance level at 10%, implying that we can statistically rule out the non-causality null hypothesis if the rolling bootstrap *p*-value is lower than this critical value, which enables us to accurately screen out time intervals with significant causal effects and further match the dynamic correlation features reflected in Figure 5. In terms of correlation direction, negative average values of the blue line represent a negative correlation between CPU and REC, whereas positive values suggest a positive correlation between the two variables. Integrating the visual trends and statistical results of the two figures, we clearly observe a prominent negative causal relationship between CPU and REC spanning from June 2014 to September 2015. Throughout this period, the gradual downward trend of overall climate policy uncertainty coincided with a sustained upward growth trend of REC, and the formal adoption and progressive implementation of the Paris Agreement in 2015 further eased policy divergence, stabilized national low-carbon development orientation and effectively buffered external policy disturbances. This empirical outcome strongly validates our earlier theoretical inference, confirming that a reduction in climate policy uncertainty can substantially facilitate and boost the scale expansion of REC. Several internal transmission mechanisms can reasonably explain this time-varying causal phenomenon. First, the decline in CPU helps optimize and refine the institutional framework and market operation mechanism of the renewable energy industry. As formal governmental policies define the required share of renewable energy within total power utilization, as well as effective linkage mechanisms connecting energy consumption structure adjustment with carbon emission quota distribution and management, market-oriented enterprises are forced to actively adjust their energy use structure and continuously raise their actual renewable energy consumption level. Second, weakened climate policy uncertainty greatly lowers the long-term policy volatility risks faced by renewable energy investment and construction projects. The officially issued 2014–2015 Energy Conservation, Emission Reduction and Low-Carbon Development Action Plan put forward concrete and quantifiable carbon emission intensity reduction goals, delivering stable, clear and forward-looking low-carbon policy signals to the whole society. Such definite policy orientation fully highlights the strategic importance and development potential of renewable energy industries; effectively consolidate market investment confidence; accelerate technological research, development and iterative upgrading in the renewable energy sector; and ultimately drive the steady growth of renewable energy consumption. Moreover, the widespread popularization and in-depth application of emerging digital technologies including smart grid dispatching systems and big data energy demand analysis can effectively alleviate serious information asymmetry existing in the energy market, cut down unnecessary transaction costs and operational management costs, and enable all market participants to respond more rationally, flexibly and adaptively to dynamic adjustments in climate policies [81]. These technological advantages further strengthen the stable investment willingness of enterprises and residential users in the renewable energy field, and significantly mitigate the negative restraining effect brought by CPU on REC.

Three specific intervals, namely, 2015:M9–2016:M3, 2017:M9–2018:M3, and 2020:M1–2020:M4, witness an obvious positive causal effect of CPU on REC, indicating that increased policy uncertainty can also facilitate renewable energy consumption under specific policy and market contexts. First, during 2015:M9–2016:M3, the rising CPU was mainly triggered by China's new climate governance commitments and strategic adjustments in the United Nations global climate framework [82]. The extensive media coverage of climate policy updates and national energy transition agendas significantly raised public and market awareness of the essential role of renewable energy in addressing climate change, prompt-

ing proactive acceptance and application of clean energy and ultimately driving the growth of REC amid heightened policy uncertainty. Second, the period of 2017:M9–2018:M3 was characterized by an overall upward CPU trend despite slight short-term fluctuations, which stemmed from frequent iterations and adjustments in domestic renewable energy policies. In 2017, China launched a pilot system for green power certificate issuance and voluntary subscription for renewable energy [83], which led market participants to form consistent expectations of gradual subsidy reduction and phase-out in the future. To fully capture the remaining policy dividends and obtain sufficient government subsidies before formal policy tightening, enterprises and investors substantially accelerated the investment and construction of renewable energy projects. Driven by this preemptive market behavior, China's total installed renewable energy capacity reached 650 million kW by December 2017, accounting for 36.6% of the country's total installed power capacity, which greatly expanded the scale of renewable energy supply and consumption. Third, the positive CPU-REC linkage during 2020:M1–2020:M4 was primarily driven by policy transition uncertainty. As 2020 was the final implementation year of China's subsidy policies for onshore wind and photovoltaic power projects, the upcoming expiration of subsidy incentives pushed market entities to rush to invest in and complete renewable energy projects to secure existing subsidy benefits before the policy shift [84], contributing to a remarkable increment of 139 million kW in national renewable energy installed capacity throughout 2020, with newly added wind and photovoltaic capacity reaching 120 million kW. Additionally, the official proposal of China's dual-carbon goals in September 2020 delivered a clear, long-term low-carbon development signal to the market [85], effectively stabilizing and boosting market confidence in the long-term development of the renewable energy industry. Even with short-term policy uncertainty arising from subsidy reforms and strategic policy adjustments, the positive long-term policy expectations strongly stimulated renewable energy investment and application, thereby further promoting the continuous growth of REC in this period.

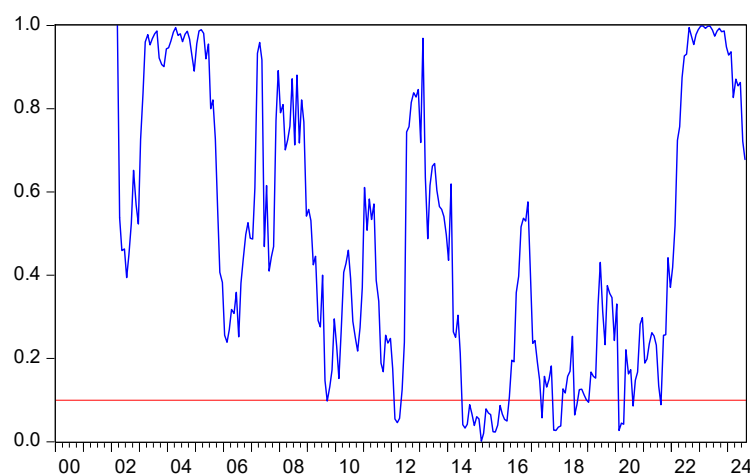


Figure 4. p -Values for the hypothesis that CPU does not Granger-cause REC. Note: Blue line = p -values for the null hypothesis “CPU does not Granger-cause REC”. Red line = 10% significance level ($p = 0.10$). Values below the red line indicate that significant Granger causality exists in a certain period.

Figure 6 plots the rolling bootstrap p -values for the null hypothesis that renewable energy consumption (REC) exerts no Granger causal effect on climate policy uncertainty (CPU), while the corresponding dynamic correlation coefficients are presented in Figure 7. Following the consistent 10% significance criterion adopted in this study, we can identify statistically significant causal intervals by examining p -value fluctuations and further

match these intervals with the time-varying correlation trends in Figure 7 to clarify the causal direction and dynamic linkage between the two variables. The results reveal significant positive causal effects of REC on CPU during two key periods, 2009:M6–2009:M12 and 2017:M6–2021:M1, demonstrating that the continuous growth of renewable energy consumption can conversely heighten climate policy uncertainty under specific development stages and energy structural contexts. During 2009:M6–2009:M12, the progressive advancement of renewable energy technologies steadily promoted the scale expansion of REC. Internationally, China made a formal commitment in 2009 to increase the proportion of renewable energy in primary energy consumption to approximately 15% by 2020. Nevertheless, the early-stage implementation of this macroscopic low-carbon target was accompanied by substantial ambiguity in specific policy implementation rules, supporting funding arrangements and financial guarantee mechanisms, which inevitably generated prominent policy uncertainty in the climate governance system [86]. With the increase in the percentage of renewable energy in the national energy structure, policymakers were required to constantly adjust and optimize relevant industrial and climate policies to further promote renewable energy deployment and accomplish established emission reduction targets, thereby further driving the increase in CPU [67]. In the longer interval of 2017:M6–2021:M1, the rapid expansion of installed photovoltaic and wind power capacity substantially boosted China's REC level. However, the booming development of the renewable energy industry triggered profound structural shocks to the traditional fossil energy sector in the short term. Driven by the rapid growth of clean energy integration, the average utilization rate of China's coal-fired power stations dropped to merely 56%, with idle capacity problems being particularly severe in the resource-rich northwest and southwest regions. The large-scale grid integration of intermittent renewable energy has squeezed the market share of traditional thermal power, resulting in massive stranded assets and considerable economic losses for the conventional energy industry [87]. Such intensified conflicts and structural contradictions between renewable energy expansion and traditional energy transformation forced the government to frequently adjust climate and energy policies to balance the coordinated development of new and old energy industries. The continuous policy revisions and trade-off adjustments ultimately induced a notable increase in climate policy uncertainty during this period.

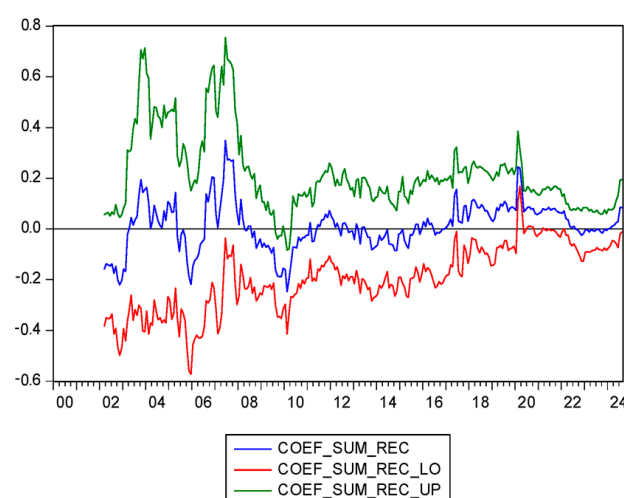


Figure 5. The coefficients for the impact of CPU on REC. Note: The blue line denotes the estimated time-varying coefficient. The red and green lines represent the 95% confidence intervals (lower and upper bounds, respectively). When the blue line takes values lower than 0, there is a negative influence of CPU on REC; otherwise, there is a positive impact.

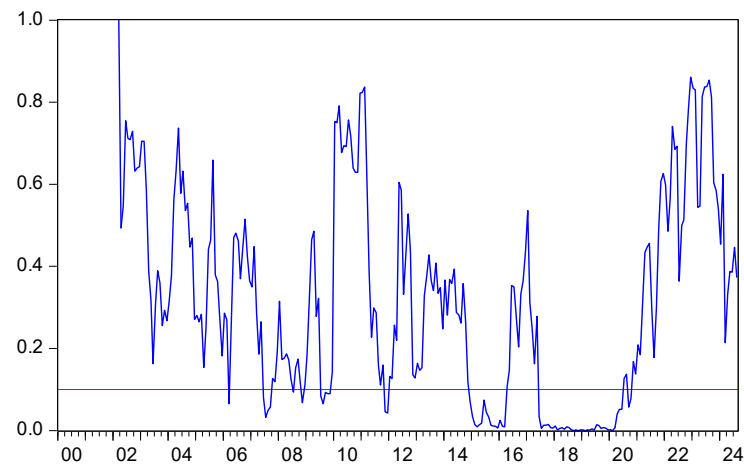


Figure 6. p -Values for the hypothesis that REC does not Granger-cause CPU. Note: Blue line = p -values for the null hypothesis “REC does not Granger-cause CPU”. Red line = 10% significance level ($p = 0.10$). Values below the red line indicate that significant Granger causality exists in a certain period.

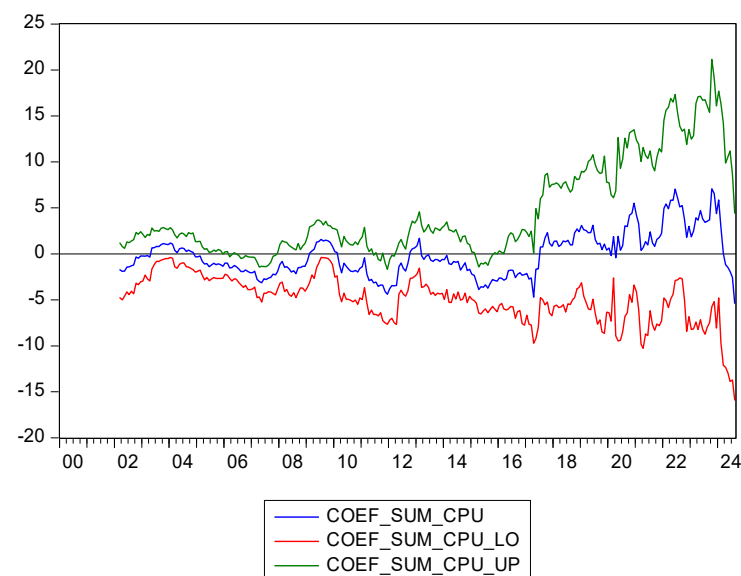


Figure 7. The coefficients for the impact of REC on CPU. Note: The blue line denotes the estimated time-varying coefficient. The red and green lines represent the 95% confidence intervals (lower and upper bounds, respectively). When the blue line takes values lower than 0, there is a negative influence of REC on CPU; otherwise, there is a positive impact.

As we can see, significant negative correlations between REC and CPU are observed during two specific intervals: 2007:M3–2007:M9 and 2015:M1–2016:M3. In both stages, the rapid development and scale expansion of renewable energy projects drove a continuous upward trend in REC, which in turn effectively reduced CPU and stabilized national low-carbon policy expectations. From March 2007 to September 2007, China’s renewable energy development entered an initial booming stage, with the annual consumption of renewable energy reaching 200 million tons of standard coal, occupying about 8% of the national total primary energy consumption, representing a 0.5% year-on-year increase. This statistical progress reflected the remarkable advancement of China’s renewable energy utilization capacity during this period. At that time, China put forward a clear phased development target, putting forward plans to elevate renewable energy’s ratio in primary energy consumption to 10% in 2010 from the initial 7%, with a target of 15% in 2020. The prominent growth momentum and huge development potential of renewable energy in

addressing climate change and optimizing the energy structure attracted sufficient attention among policymakers. To standardize industrial development and achieve the established low-carbon goals, the China launched the National Plan for Developing Renewable Energy, which prioritized the large-scale development and utilization of hydropower and wind power industries [44]. Driven by clear industrial development planning and definite policy orientation, the government formed stable, long-term and systematic climate and energy policy arrangements for the renewable energy sector. The standardized and forward-looking policy system effectively avoided frequent policy adjustments and institutional uncertainties, thereby significantly reducing the overall level of CPU. For the period of 2015:M1–2016:M3, the negative REC–CPU linkage was closely related to China’s large-scale investment layout and the industrial upgrading of renewable energy during the 13th Five-Year Plan period, with a total investment of approximately 2.3 trillion yuan injected into the renewable energy industry. Benefiting from strong policy support and capital input, China’s renewable energy industry, particularly the new energy vehicle sector, achieved explosive growth in 2015. The annual output and sales of new energy vehicles reached 340,471 and 331,091 units respectively, both surging to more than four times the previous year’s volume. Rising popularity of electric vehicles facilitates the coordinated construction of relevant charging infrastructures, forming a synergistic development pattern with renewable energy power generation and grid connection [88]. The integrated development of the energy and transportation industries promoted cross-departmental communication and coordinated policy formulation, effectively eliminating climate policy uncertainty caused by inter-sectoral policy fragmentation, misalignment and inconsistent regulatory standards. Overall, the sustained growth of REC in the above two stages facilitated the optimization, standardization and stabilization of China’s climate and energy policy system, ultimately lowering climate policy uncertainty. Accordingly, these empirical findings fully verify the validity of Hypothesis 3 (H3) proposed in this study.

6. Discussion and Conclusions

6.1. Discussion

This paper examines the dynamic causality from CPU to REC in order to better realize the interaction process between China’s climate policy and renewable energy. A rolling window causality test is utilized to determine that these two variables are causally related periods, with a sample period of 2000:M1–2024:M8, which can address the limitations of the time-series linearity assumption. The results indicate that the causal link from CPU to REC differs across various intervals. First, CPU almost always plays a promoting role in REC, whether it is a positive or negative causal effect. Second, there seems to be no doubt that increased REC will mitigate CPU. REC can decrease climate policy risks by making governments see its potential in addressing climate change. However, the impact of the swift penetration of renewable energy into the traditional energy industry cannot be ignored. The findings offer an empirical causal inference basis for managing CPU, particularly in renewable energy, and provide new insights to deepen the understanding of REC drivers and energy transition dynamics.

6.2. Conclusions

Drawing on the empirical findings of the time-varying and asymmetric CPU-REC nexus and the moderating role of digitalization and technological innovation, this study puts forward targeted policies for China’s low-carbon energy transition and dual-carbon governance.

First, given the time-varying positive CPU-REC relationship, China should build a stable and long-term climate policy framework. The results demonstrate that CPU facili-

tates renewable energy consumption, yet frequent fragmented policy adjustments induce industrial resistance and transition risks. To advance the dual-carbon goals, policymakers should abandon short-term regulatory interventions and adopt a transparent, long-term energy transition roadmap. Setting binding renewable energy targets and improving policy expectation mechanisms can strengthen positive policy incentives and stabilize China's energy market restructuring.

Second, due to the asymmetric REC feedback on CPU, China should coordinate traditional and renewable energy development to resolve structural conflicts. This asymmetric nexus matches China's dual energy structure of coexisting thermal and renewable power. Instead of pursuing simplistic energy replacement, China should adopt a coordinated transition model. Optimizing renewable portfolio standards and green certificate trading and launching a just energy transition fund can balance the interests of traditional and new energy sectors, supporting steady low-carbon transformation.

Third, with the enabling effects of digitalization and technological innovation, China can enhance the resilience of its energy transition. Digital and technological innovation effectively moderate the CPU-REC relationship and mitigate policy uncertainty risks. China should promote intelligent grid dispatching, blockchain-based green certificate trading for distributed renewables, and differentiated R&D subsidy schemes. These technology-oriented measures hedge policy shocks, stabilize market expectations, and foster an efficient and sustainable low-carbon energy system.

While this paper conducts an analysis of CPU and REC in China, several limitations remain, which point out promising avenues for future research. First, restricted by data accessibility, official statistics on China's actual renewable energy consumption are yet to be publicly released. Accordingly, this study adopts new energy generation as a proxy variable instead. Nevertheless, this alternative indicator may generate measurement biases stemming from power transmission losses and other practical factors. Hence, subsequent research can incorporate precise actual renewable energy consumption data to correct such estimation deviations once relevant official data become available. Second, this research study focuses solely on China given its unique institutional and energy characteristics, failing to account for cross-national disparities between developed and developing economies. In further studies, we will construct climate policy uncertainty indices for various economies and employ panel data models to explore the heterogeneous impacts across different countries.

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